WORKED SOLUTION

STEP II 1987, Question 8

Pure • Vector Loci in 3D

STEP II 1987, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q8

The key idea

Each equation becomes recognisable once it is written in terms of distances from the points A and B with position vectors $\mathbf{a}$ and $\mathbf{b}$. For the equivalence, everything can be expressed using just two numbers, $P = |\mathbf{r}-\mathbf{a}|^2$ and $Q = |\mathbf{r}-\mathbf{b}|^2$. Then (i) and (ii) become two simple linear equations in P and Q, and so do (iii) and (iv).

Solution

Throughout, A, B and R are the points with position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{r}$, and M is the midpoint of AB, with position vector $\mathbf{m} = \frac{1}{2}(\mathbf{a}+\mathbf{b})$. We assume $\mathbf{a} \neq \mathbf{b}$ (the case $\mathbf{a} = \mathbf{b}$ is dealt with at the end), and write $d = |\mathbf{a}-\mathbf{b}| = AB$.

Part (i)

Expanding,

$$\begin{aligned}|\mathbf{r}-\mathbf{a}|^2 - |\mathbf{r}-\mathbf{b}|^2 &= (|\mathbf{r}|^2 - 2\mathbf{a}\cdot\mathbf{r} + |\mathbf{a}|^2) - (|\mathbf{r}|^2 - 2\mathbf{b}\cdot\mathbf{r} + |\mathbf{b}|^2) \\ &= -2\Big[(\mathbf{a}-\mathbf{b})\cdot\mathbf{r} - \tfrac{1}{2}(|\mathbf{a}|^2 - |\mathbf{b}|^2)\Big].\end{aligned}$$

So (i) holds exactly when $|\mathbf{r}-\mathbf{a}| = |\mathbf{r}-\mathbf{b}|$, that is, when $RA = RB$. The locus is the **plane that perpendicularly bisects** AB: the plane through M with normal $\mathbf{a}-\mathbf{b}$. (We can see it is a plane directly, since (i) has the form $\mathbf{n}\cdot\mathbf{r} = \text{constant}$ with $\mathbf{n} = \mathbf{a}-\mathbf{b} \neq \mathbf{0}$, and $\mathbf{r} = \mathbf{m}$ satisfies it because $(\mathbf{a}-\mathbf{b})\cdot(\mathbf{a}+\mathbf{b}) = |\mathbf{a}|^2 - |\mathbf{b}|^2$.)

Part (ii)

Equation (ii) says that $\overrightarrow{RA}$ and $\overrightarrow{RB}$ are perpendicular, or that R is A or B. To identify the locus, complete the square:

$$\begin{aligned}|\mathbf{r}-\mathbf{m}|^2 &= |\mathbf{r}|^2 - (\mathbf{a}+\mathbf{b})\cdot\mathbf{r} + \tfrac{1}{4}|\mathbf{a}+\mathbf{b}|^2 \\ &= (\mathbf{a}-\mathbf{r})\cdot(\mathbf{b}-\mathbf{r}) - \mathbf{a}\cdot\mathbf{b} + \tfrac{1}{4}|\mathbf{a}+\mathbf{b}|^2 \\ &= (\mathbf{a}-\mathbf{r})\cdot(\mathbf{b}-\mathbf{r}) + \tfrac{1}{4}|\mathbf{a}-\mathbf{b}|^2.\end{aligned}$$

So (ii) is equivalent to $|\mathbf{r}-\mathbf{m}| = \frac{1}{2}d$. The locus is the **sphere with AB as a diameter**: centre M, radius $\frac{1}{2}AB$. This is the three-dimensional version of "the angle in a semicircle is a right angle".

Parts (iii) and (iv)

Equation (iii) says $RA = d/\sqrt{2}$. Its locus is the **sphere with centre A and radius $AB/\sqrt{2}$**. In the same way, the locus of (iv) is the **sphere with centre B and radius $AB/\sqrt{2}$**.

The equivalence

Let $P = |\mathbf{r}-\mathbf{a}|^2$ and $Q = |\mathbf{r}-\mathbf{b}|^2$. Part (i) showed that

$$\text{(i)} \iff P = Q.$$

For (ii), write $\mathbf{a}-\mathbf{b} = (\mathbf{a}-\mathbf{r}) - (\mathbf{b}-\mathbf{r})$ and square:

$$d^2 = |\mathbf{a}-\mathbf{r}|^2 + |\mathbf{b}-\mathbf{r}|^2 - 2(\mathbf{a}-\mathbf{r})\cdot(\mathbf{b}-\mathbf{r}) = P + Q - 2(\mathbf{a}-\mathbf{r})\cdot(\mathbf{b}-\mathbf{r}).$$

Hence

$$\text{(ii)} \iff P + Q = d^2.$$

Equations (iii) and (iv) say $P = \frac{1}{2}d^2$ and $Q = \frac{1}{2}d^2$. Now

$$(P = Q \text{ and } P + Q = d^2) \iff P = Q = \tfrac{1}{2}d^2.$$

The forward direction holds because substituting $Q = P$ gives $2P = d^2$. For the reverse, if $P = Q = \frac{1}{2}d^2$ then certainly $P = Q$ and $P + Q = d^2$. Every step above is an "if and only if", so (i) and (ii) together are equivalent to (iii) and (iv) together, as required.

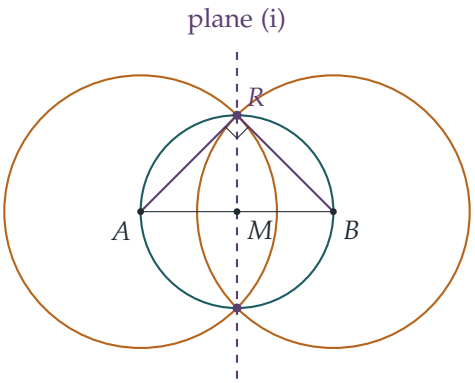
> **Ask yourself** How can a plane meeting a sphere give the same set as two spheres meeting?

Both pairs describe the same circle. The plane (i) passes through M, the centre of the sphere (ii), so it cuts that sphere in a great circle: centre M, radius $\frac{1}{2}AB$, lying in the perpendicular bisector plane of AB. The spheres (iii) and (iv) have equal radii $AB/\sqrt{2}$, and these sum to more than AB, so they do meet. By symmetry they meet in a circle in the perpendicular bisector plane, centred at M. Its radius is $\sqrt{\frac{1}{2}AB^2 - \frac{1}{4}AB^2} = \frac{1}{2}AB$, by Pythagoras in triangle AMR, so it is the same circle.

Geometrical meaning

A point R lies on this circle exactly when triangle ARB is *right-angled and isosceles*, with the right angle at R. The pair (i) and (ii) describes this through its properties: $RA = RB$ (isosceles) and $\angle ARB = 90^\circ$ (right angle). The pair (iii) and (iv) describes it through its side lengths: in a right isosceles triangle with hypotenuse AB, both legs have length $AB/\sqrt{2}$. The algebra above is Pythagoras' theorem and its converse in vector form. The identity $d^2 = P + Q - 2(\mathbf{a} - \mathbf{r}) \cdot (\mathbf{b} - \mathbf{r})$ is the cosine rule in triangle ARB.

The diagram shows a cross-section by any plane containing A and B. The bisecting plane appears as the dashed line, the sphere (ii) as the teal circle, and the spheres (iii) and (iv) as the amber circles. All four meet at the two points where the circle of solutions crosses the page.



The degenerate case

If $\mathbf{a} = \mathbf{b}$, then (i) reads $0 = 0$ and holds for all $\mathbf{r}$, while (ii), (iii) and (iv) each say $\mathbf{r} = \mathbf{a}$. The equivalence still holds, but every locus collapses to the single point A (or to all of space for (i)).

Checks

Take $\mathbf{a} = (1, 0, 0)$ and $\mathbf{b} = (-1, 0, 0)$. Then (i) is the plane $x = 0$, and (ii) is $x^2 + y^2 + z^2 = 1$. Equations (iii) and (iv) are $(x \mp 1)^2 + y^2 + z^2 = 2$, and subtracting them gives $x = 0$ again. All four meet in the unit circle $y^2 + z^2 = 1$, $x = 0$. For example, $R = (0, 1, 0)$ gives $RA = RB = \sqrt{2}$ and $\overrightarrow{RA} \cdot \overrightarrow{RB} = -1 + 1 = 0$. We also checked all the identities above exactly, in rational arithmetic, for 5000 random choices of $\mathbf{a}$, $\mathbf{b}$, $\mathbf{r}$. We confirmed that random points on the predicted circle satisfy all four equations.

Common slips

- Describing (ii) as "a circle". In three dimensions it is a sphere, and the locus of (i) is a plane, not a line.

- Proving only one direction. The question asks for an equivalence, so make sure each step reverses, or prove both directions.
- Saying the two pairs "both give a circle" without checking it is the *same* circle: same plane, same centre, same radius.

Beyond STEP

Each of these loci is a level set of a quadratic function of $\mathbf{r}$, and the trick of subtracting two sphere equations to get a plane is the idea of the *radical plane* of two spheres. More generally, describing the same geometric object by different systems of equations is the starting point of algebraic geometry. Here, a circle in space is cut out by two equations, and many different pairs of equations do the job. Strang's *Introduction to Linear Algebra* is a good place to strengthen the vector algebra, in particular the link between dot products, lengths and angles.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.